

Observation Refinement in Propositional Logical Topologies

Fiber profiles, adjoint criteria, Quillen fibers, refinement defects, and the limits of complete classification

Working research note v0.2 — 23 September 2026

Follow-up to *Homological Foundations for Propositional Logical Topologies*, built on the ProLT core manuscript v0.8.

Status convention. [S] denotes standard finite-poset/algebraic-topological mathematics. [A] denotes a direct ProLT specialization or elementary theorem derived from the v0.8 signature framework. [C] denotes a potentially publishable ProLT-specific formulation, invariant, or algorithmic consequence whose novelty should be subjected to a dedicated literature audit before any priority claim.

1 Question and answer in one page

Let $\Phi = (\phi_i)_{i \in I}$ be an observation family and let Ψ extend Φ by new observations. The v0.8 ProLT theory gives the realized-signature posets

$$P_\Phi = o_\Phi(\Omega_P), \quad P_\Psi = o_\Psi(\Omega_P),$$

ordered by inclusion, and a canonical forgetful refinement map

$$r_{\Psi\Phi} : P_\Psi \longrightarrow P_\Phi$$

obtained by deleting the new coordinates. The corresponding order complexes are

$$O_\Phi = \Delta(P_\Phi), \quad O_\Psi = \Delta(P_\Psi).$$

The central problem is:

When does adding observations change only the resolution of the presentation, and when does it change the homotopy or homology of the observable-state space?

The main results of this note give a hierarchy of answers.

- R1.** Every refinement admits an exact *fiber-profile normal form*: over each coarse signature p , record exactly which new-coordinate signatures occur. This completely reconstructs P_Ψ and the map $r_{\Psi\Phi}$.
- R2.** The refinement is *topologically redundant* exactly when every coarse state has one new profile and those profiles are monotone. Equivalently, $r_{\Psi\Phi}$ is an order isomorphism.
- R3.** If the refinement does not split coarse states, then O_Ψ is literally a subcomplex of O_Φ . A coarse chain survives exactly when its new-coordinate profiles increase along the chain. For one new Boolean observation, the deleted chains are precisely those containing a $1 \rightarrow 0$ descent.
- R4.** There are simple signature-level sufficient criteria for homotopy invariance. If every lower refinement fiber has a greatest element, or every upper refinement fiber has a least element, then $r_{\Psi\Phi}$ has an order adjoint and is a homotopy equivalence. For one observation ψ , this says: *possibility of ψ* or *forced truth of ψ* must be monotone in the coarse information order.

- R5.** More generally, Quillen–McCord/Barmak fiber criteria apply. Contractible principal fibers imply a simple homotopy equivalence; acyclic principal fibers imply homology equivalence.
- R6.** Homology equivalence has an exact finite test: the refinement mapping cone is acyclic. In the non-splitting case it reduces to ordinary relative homology $H_*(O_\Phi, O_\Psi)$.
- R7.** There is no unrestricted algorithmic classification of homotopy-neutral refinements: arbitrary finite posets can be realized as refined signature posets above a one-point coarse space, so the problem contains finite-complex contractibility, which is undecidable. Homology neutrality, by contrast, remains decidable by finite boundary-matrix computation.

2 Base ProLT data and refinement map

Let $P = \{p_1, \dots, p_n\}$ be a finite propositional variable set and $\Omega_P = \{0, 1\}^P$ its valuation space. For a formula L , write

$$\zeta(L) = \{v \in \Omega_P : v \models L\}.$$

For an indexed observation family $\Phi = (\phi_i)_{i \in I}$, define

$$o_\Phi(v) = \{i \in I : v \models \phi_i\} \subseteq I.$$

The v0.8 ProLT results identify topological indistinguishability with equality of signatures and the specialization preorder with inclusion:

$$v \sim_\Phi w \iff o_\Phi(v) = o_\Phi(w), \quad v \preceq_\Phi w \iff o_\Phi(v) \subseteq o_\Phi(w).$$

Hence the Kolmogorov quotient is the finite poset

$$P_\Phi = o_\Phi(\Omega_P) \subseteq 2^I$$

with inclusion order.

Suppose now that

$$\Psi = (\Phi, \Theta)$$

extends Φ by a disjointly indexed family $\Theta = (\theta_a)_{a \in A}$. Then

$$o_\Psi(v) = (o_\Phi(v), o_\Theta(v)) \in 2^I \times 2^A,$$

and the coordinate-forgetting map

$$r = r_{\Psi\Phi} : P_\Psi \rightarrow P_\Phi, \quad r(s) = s \cap I,$$

is surjective and monotone.

3 The fiber-profile normal form

Definition 3.1 (Coarse state and refinement profile). For $p \in P_\Phi$, let

$$C_p := o_\Phi^{-1}(p)$$

be the coarse observational state represented by p . Define its *new-coordinate fiber profile*

$$\mathcal{B}_p := o_\Theta(C_p) = \{o_\Theta(v) : v \in C_p\} \subseteq 2^A.$$

Each $\mathcal{B}_p$ is nonempty.

Theorem 3.2 (Fiber-profile normal form). *[A]/[C] Let*

$$Q_{\mathcal{B}} := \{(p, a) : p \in P_{\Phi}, a \in \mathcal{B}_p\} \subseteq P_{\Phi} \times 2^A$$

with product order

$$(p, a) \leq (q, b) \iff p \leq q \text{ in } P_{\Phi} \text{ and } a \subseteq b.$$

Then

$$P_{\Psi} \cong Q_{\mathcal{B}}$$

as posets, under separation of old and new coordinates. Under this isomorphism the refinement map is exactly the first-coordinate projection

$$r(p, a) = p.$$

Consequently the family $(\mathcal{B}_p)_{p \in P_{\Phi}}$ determines the refined signature poset and the refinement projection completely.

Proof. Every realized Ψ -signature comes from a valuation v and splits uniquely as

$$(o_{\Phi}(v), o_{\Theta}(v)) = (p, a)$$

with $a \in \mathcal{B}_p$. Conversely, every pair in $Q_{\mathcal{B}}$ is realized by the definition of $\mathcal{B}_p$. Inclusion of the combined signatures is coordinatewise inclusion because the old and new index sets are disjoint. The projection statement is immediate. $\square$

Remark 3.3 (Point fibers are not Quillen fibers). The direct point fiber

$$r^{-1}(p) = \{p\} \times \mathcal{B}_p$$

records only how the coarse state p splits. Quillen-type theorems use instead the preimages of principal ideals or filters,

$$r^{-1}(\downarrow p), \quad r^{-1}(\uparrow p),$$

which also remember how the split pieces compare with states below or above p . This distinction is essential: every point fiber can be contractible while the refinement still changes homotopy type.

4 Exact classification of topological redundancy

The v0.8 manuscript calls a new observation topologically redundant when its truth region is already open in the old ProLT. The fiber normal form turns this into an exact signature-level theorem for an arbitrary finite block of added observations.

Theorem 4.1 (Redundancy/order-isomorphism theorem). *[A]/[C] For a refinement $\Psi = (\Phi, \Theta)$, the following are equivalent.*

- (i) $\tau_{\Psi} = \tau_{\Phi}$ on the fixed valuation carrier Ω_P .
- (ii) Every new truth region $\varsigma(\theta_a)$, $a \in A$, is already τ_{Φ} -open.

(iii) Every fiber profile is a singleton,

$$\mathcal{B}_p = \{\alpha(p)\},$$

and the resulting map

$$\alpha : P_\Phi \rightarrow 2^A$$

is isotone:

$$p \leq q \implies \alpha(p) \subseteq \alpha(q).$$

(iv) The refinement projection $r : P_\Psi \rightarrow P_\Phi$ is an order isomorphism.

Proof. The equivalence of (i) and (ii) is the definition of equality of the generated topologies: $\tau_\Phi \subseteq \tau_\Psi$ automatically, and equality holds exactly when each new subbasic truth region is old-open.

Assume (i). If two valuations lie in the same old signature class C_p , they are topologically indistinguishable in τ_Φ ; equality of topologies makes them indistinguishable in τ_Ψ as well. Thus the new signature is constant on C_p , giving $\mathcal{B}_p = \{\alpha(p)\}$. If $p \leq q$, choose $v \in C_p$, $w \in C_q$. The old specialization relation gives $v \preceq_\Phi w$. Equality of topologies gives $v \preceq_\Psi w$, hence $\alpha(p) \subseteq \alpha(q)$. Therefore (iii) holds.

Assume (iii). For each new coordinate $a \in A$, the set

$$U_a := \{p \in P_\Phi : a \in \alpha(p)\}$$

is upward closed. By the signature representation of the ProLT topology,

$$\zeta(\theta_a) = o_\Phi^{-1}(U_a)$$

is τ_Φ -open. Hence (ii).

Under (iii), Theorem 3.2 identifies P_Ψ with the graph $\{(p, \alpha(p))\}$. Monotonicity of α says

$$p \leq q \iff (p, \alpha(p)) \leq (q, \alpha(q)),$$

so projection is an order isomorphism. Conversely, if projection is an order isomorphism then every fiber has one point and its inverse is necessarily isotone, giving (iii). $\square$

Remark 4.2. This theorem separates three notions which should not be conflated:

- (a) *no splitting*: each $\mathcal{B}_p$ is a singleton;
- (b) *no order deletion*: the singleton profile map is isotone;
- (c) *full redundancy*: both hold, equivalently the topology is unchanged.

A refinement can leave every coarse equivalence class intact while still deleting one-way specialization comparisons.

5 Non-splitting refinements and chain deletion

Definition 5.1 (Non-splitting refinement). A refinement is *non-splitting* if every $\mathcal{B}_p$ is a singleton. Write

$$\mathcal{B}_p = \{\alpha(p)\}.$$

No monotonicity of α is assumed.

Theorem 5.2 (Order deletion theorem). *[A]/[C] For a non-splitting refinement, identify the fine and coarse vertex sets through the bijection r . Then*

$$p \leq_\Psi q \iff p \leq_\Phi q \text{ and } \alpha(p) \subseteq \alpha(q).$$

Consequently

$$\leq_\Psi \subseteq \leq_\Phi$$

and

$$O_\Psi \subseteq O_\Phi$$

as a simplicial subcomplex on the same vertex set. A coarse chain

$$p_0 < p_1 < \dots < p_k$$

survives the refinement exactly when

$$\alpha(p_0) \subseteq \alpha(p_1) \subseteq \dots \subseteq \alpha(p_k).$$

Proof. In the fiber-profile normal form the unique fine point over p is $(p, \alpha(p))$. Product order gives the displayed criterion. Every fine chain is therefore a coarse chain; conversely, a coarse chain is fine exactly when its new profiles are coordinatewise nondecreasing. $\square$

Corollary 5.3 (One-bit descent criterion). *Suppose one new Boolean observation ψ is constant on each coarse class, and write*

$$b(p) \in \{0, 1\}$$

for its value on C_p . Then a coarse chain $p_0 < \dots < p_k$ survives exactly when

$$b(p_0) \leq b(p_1) \leq \dots \leq b(p_k).$$

Equivalently, it is deleted exactly when it contains a $1 \rightarrow 0$ inversion.

Corollary 5.4 (Post- T_0 refinement filtration). *[A]/[C] If X_Φ is already T_0 on the valuation carrier, then o_Φ is injective, so every later observation refinement is automatically non-splitting. Thus a sequence*

$$\Phi_0 \subseteq \Phi_1 \subseteq \dots \subseteq \Phi_T$$

with $X_{\Phi_0} T_0$ yields, after identifying vertices with valuations,

$$O_{\Phi_T} \subseteq \dots \subseteq O_{\Phi_1} \subseteq O_{\Phi_0}.$$

Hence ordinary persistence can be applied after reversing the refinement index; zigzag persistence is not required while observations are only added.

Proof. A coarse signature class contains a single valuation when o_Φ is injective, so it cannot split further as a set. Theorem 5.2 then applies at every later stage. $\square$

6 Relative defect chains in the non-splitting case

The preceding inclusion lets the mapping-cone defect from the foundation note be sharpened.

Theorem 6.1 (Refinement defect becomes relative homology). *[S]/[A] For a non-splitting refinement, under the vertex identification of Theorem 5.2, the induced simplicial map is the inclusion*

$$O_\Psi \hookrightarrow O_\Phi.$$

Therefore the mapping-cone refinement defect satisfies

$$D_k(\Psi/\Phi; R) \cong H_k(O_\Phi, O_\Psi; R).$$

Definition 6.2 (Refinement-descent chain complex). For a non-splitting refinement define $C_k^{\text{desc}}(\Psi/\Phi; R)$ to be the free R -module generated by the coarse k -chains that fail the survival condition

$$\alpha(p_0) \subseteq \cdots \subseteq \alpha(p_k).$$

Its boundary is the ordinary simplicial boundary followed by deletion of every face that does survive the refinement.

Proposition 6.3 (Exact descent-chain computation). *[A]/[C] There is a natural chain isomorphism*

$$C_*^{\text{desc}}(\Psi/\Phi; R) \cong C_*(O_\Phi, O_\Psi; R).$$

Hence

$$D_k(\Psi/\Phi; R) \cong H_k(C_*^{\text{desc}}(\Psi/\Phi; R)).$$

For one added Boolean observation, the generators are exactly coarse chains containing a $1 \rightarrow 0$ descent.

Proof. The simplicial chain group of O_Φ is free on its simplices, and the subcomplex O_Ψ is spanned by exactly the surviving simplices. The quotient chain group therefore has the nonsurviving simplices as a basis. Its differential is the coarse simplicial differential modulo the surviving submodule, which is precisely the stated rule. $\square$

7 Adjoint criteria: an elementary homotopy-neutral class

For $p \in P_\Phi$, define the principal refinement fibers

$$L_p := r^{-1}(\downarrow p), \quad U_p := r^{-1}(\uparrow p).$$

The following test is much easier than computing arbitrary homotopy types.

Theorem 7.1 (Greatest-lower-fiber / least-upper-fiber criterion). *[S]/[A]/[C] Let*

$$U(p) := \bigcup_{\substack{q \leq p \\ a \in \mathcal{B}_q}} a, \quad L(p) := \bigcap_{\substack{q \geq p \\ a \in \mathcal{B}_q}} a.$$

Then:

(i) L_p has a greatest element if and only if

$$\mathbf{U}(p) \in \mathcal{B}_p.$$

When this holds, the greatest element is $(p, \mathbf{U}(p))$.

(ii) U_p has a least element if and only if

$$\mathbf{L}(p) \in \mathcal{B}_p.$$

When this holds, the least element is $(p, \mathbf{L}(p))$.

If (i) holds for every p , then r has a right adjoint. If (ii) holds for every p , then r has a left adjoint. Either global condition implies that

$$\Delta(r) : \mathcal{O}_\Psi \rightarrow \mathcal{O}_\Phi$$

is a homotopy equivalence.

Proof. Suppose L_p has greatest element (q, a_*) . Since r is surjective, choose some $(p, b) \in r^{-1}(p) \subseteq L_p$. Greatness gives $(p, b) \leq (q, a_*)$, so $p \leq q \leq p$ and therefore $q = p$. The second coordinate a_* contains every new signature occurring over every $q \leq p$, hence contains $\mathbf{U}(p)$. Since a_* itself occurs in the union, equality follows. Conversely, if $\mathbf{U}(p) \in \mathcal{B}_p$, then every $(q, a) \in L_p$ satisfies

$$q \leq p, \quad a \subseteq \mathbf{U}(p),$$

so $(p, \mathbf{U}(p))$ is greatest. This proves (i). The proof of (ii) is dual.

If every lower fiber has a greatest element, define

$$s_R(p) := (p, \mathbf{U}(p)).$$

Then

$$r(x) \leq p \iff x \leq s_R(p),$$

so $r \dashv s_R$. In particular $rs_R = \text{id}$ and

$$\text{id} \leq s_R r.$$

Order-preserving maps that are pointwise comparable induce homotopic maps on order complexes, so $\Delta(s_R)$ is a homotopy inverse to $\Delta(r)$. The left-adjoint case is dual. $\square$

7.1 A sharp one-observation interpretation

Let ψ be one newly added observation. For each coarse state C_p , define

$$M_\psi(p) = \begin{cases} 1, & \text{some } v \in C_p \text{ satisfies } \psi, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$m_\psi(p) = \begin{cases} 1, & \text{every } v \in C_p \text{ satisfies } \psi, \\ 0, & \text{otherwise.} \end{cases}$$

Thus M_ψ records *possibility* of the new observation inside a coarse state, while m_ψ records *forced truth*. The possible fiber profiles are

$$\{0\}, \quad \{1\}, \quad \{0, 1\},$$

with

$$m_\psi(p) = \min \mathcal{B}_p, \quad M_\psi(p) = \max \mathcal{B}_p.$$

Corollary 7.2 (Possible/forced monotonicity criterion). *[A]/[C] For a one-observation refinement:*

- (i) *r has a right adjoint if and only if $M_\psi : P_\Phi \rightarrow \{0, 1\}$ is isotone.*
- (ii) *r has a left adjoint if and only if $m_\psi : P_\Phi \rightarrow \{0, 1\}$ is isotone.*

Equivalently, either of the following is sufficient for homotopy invariance:

- *the coarse states in which ψ is possible form an upset;*
- *the coarse states in which ψ is forced form an upset.*

Topological redundancy is the stronger special case

$$m_\psi = M_\psi = b$$

with b isotone.

Proof. For one bit,

$$U(p) = \max_{q \leq p} M_\psi(q).$$

The membership condition $U(p) \in \mathcal{B}_p$ is equivalent to $U(p) = M_\psi(p)$, which for every p is exactly isotonicity of M_ψ . Dually,

$$L(p) = \min_{q \geq p} m_\psi(q)$$

and $L(p) \in \mathcal{B}_p$ for every p is exactly isotonicity of m_ψ . □

8 Quillen–McCord/Barmak fiber criteria

The adjoint criterion forces a cone point in every principal fiber. Quillen’s fiber theorem permits much more general contractible fibers.

Theorem 8.1 (Contractible-fiber criterion). *[S]/[A] If for every $p \in P_\Phi$, the order complex*

$$\Delta(r^{-1}(\downarrow p))$$

is contractible, then

$$\Delta(r) : O_\Psi \longrightarrow O_\Phi$$

is a simple homotopy equivalence, hence a homotopy equivalence. Dually, the same conclusion follows if every

$$\Delta(r^{-1}(\uparrow p))$$

is contractible.

Proof. The lower-fiber statement is the finite-poset McCord–Quillen theorem together with Barmak’s strengthening to simple homotopy equivalence. The upper-fiber statement follows by applying the theorem to opposite posets. □

Theorem 8.2 (Acyclic-fiber homology criterion). *[S]/[A] Fix coefficients R . If*

$$\tilde{H}_i(\Delta(r^{-1}(\downarrow p)); R) = 0$$

for every $p \in P_\Phi$ and every $i \geq 0$, then

$$r_* : H_i(O_\Psi; R) \longrightarrow H_i(O_\Phi; R)$$

is an isomorphism for every i . The dual upper-fiber statement also holds.

Remark 8.3 (Sufficient, not necessary). The principal-fiber conditions are powerful but not necessary. Section 11 gives a concrete ProLT refinement for which r is a homotopy equivalence even though one lower principal fiber and one upper principal fiber are each two disconnected points. Thus neither orientation of the Quillen test succeeds, despite homotopy neutrality.

9 Exact homology classification by the refinement cone

The homotopy problem is difficult in full generality, but the homology problem has an exact algebraic answer.

Definition 9.1 (Refinement mapping cone). Let

$$r_\# : C_*(O_\Psi; R) \longrightarrow C_*(O_\Phi; R)$$

be the induced chain map. Define

$$D_*(\Psi/\Phi; R) = \text{Cone}(r_\#)$$

and

$$D_k(\Psi/\Phi; R) = H_k(D_*(\Psi/\Phi; R)).$$

Theorem 9.2 (Exact homology-equivalence criterion). *[S]/[A]/[C] The following are equivalent:*

(i) $r_* : H_k(O_\Psi; R) \rightarrow H_k(O_\Phi; R)$ is an isomorphism for every k ;

(ii)

$$D_k(\Psi/\Phi; R) = 0 \quad \text{for every } k.$$

Thus the refinement cone gives an exact, finite homological classification of whether the refinement is homology-neutral.

Proof. The standard mapping-cone long exact sequence is

$$\cdots \rightarrow H_k(O_\Psi) \xrightarrow{r_*} H_k(O_\Phi) \rightarrow D_k(\Psi/\Phi) \rightarrow H_{k-1}(O_\Psi) \rightarrow \cdots.$$

Exactness gives the equivalence. □

Corollary 9.3 (Quick obstructions). *If any of the following differs between O_Ψ and O_Φ , the refinement is not homology-neutral:*

- (a) a Betti number over the chosen field;
- (b) the number of connected components;
- (c) the Euler characteristic.

The converses are false in general.

9.1 An exact H_0 criterion

Proposition 9.4 (Connected-component criterion). *[A] Let $\pi_0(P)$ denote connected components of the undirected comparability graph of a finite poset P . Since r is surjective, the induced map*

$$r_* : H_0(O_\Psi; R) \rightarrow H_0(O_\Phi; R)$$

is always surjective. It is an isomorphism if and only if, for every connected component $C \subseteq P_\Phi$, the inverse image

$$r^{-1}(C)$$

is connected.

Proof. A connected component of P_Ψ maps into one component of P_Φ because comparability paths map to comparability-or-equality paths. Surjectivity of r ensures every coarse component receives at least one fine component. On H_0 , each fine component generator maps to the generator of its coarse component. The map is injective exactly when no coarse component receives more than one fine component. $\square$

10 A strict hierarchy of refinement-neutrality tests

It is useful to distinguish five levels.

Class	Condition
R	r is an order isomorphism (topological redundancy)
A	r has a left or right order adjoint
Q	all lower or all upper principal refinement fibers are contractible
H	$\Delta(r)$ is a homotopy equivalence
M	r_* is a homology isomorphism

Standard implications and the preceding theorems give

$$\boxed{R \subseteq A \subseteq Q \subseteq H \subseteq M.}$$

The examples below show that the first three inclusions are already strict in very small Boolean ProLTs. The universality result of Section 13 shows that $H \subsetneq M$ also occurs in unrestricted ProLT refinements by importing any finite acyclic noncontractible complex.

11 Worked Boolean examples

Throughout, signatures are ordered by inclusion and coefficients for displayed homology groups are $\mathbb{F}_2$.

11.1 Redundant: adding $X \wedge Y$ to (X, Y)

Let

$$\Phi = (X, Y), \quad \psi = X \wedge Y.$$

The coarse signatures are

$$D = \emptyset, \quad A = \{X\}, \quad C = \{Y\}, \quad B = \{X, Y\},$$

forming the Boolean diamond. The new bit is

$$b(D) = b(A) = b(C) = 0, \quad b(B) = 1.$$

It is isotone, so Theorem 4.1 applies. The refinement is in $\mathbf{R}$: the ProLT topology and canonical order complex are unchanged, although a weighted signature metric may change if the redundant coordinate is assigned positive weight.

11.2 Adjoint-neutral but nonredundant: adding X to no observations

Take one propositional variable X , let Φ be the empty observation family, and refine by $\Psi = (X)$. The coarse poset is a point; the fine poset is the two-element chain

$$0 < 1.$$

The projection to a point has both a left adjoint (choose the minimum) and a right adjoint (choose the maximum), so it is a homotopy equivalence. The topology nevertheless changes from indiscrete to Sierpiński. Thus this refinement lies in $\mathbf{A} \setminus \mathbf{R}$.

11.3 Quillen-neutral but no adjoint: adding XOR to (X, Y)

Let

$$\Phi = (X, Y), \quad \psi = X \updownarrow Y.$$

The new-bit profile on (D, A, C, B) is

$$(0, 1, 1, 0).$$

No coarse class splits, but the old comparisons $A < B$ and $C < B$ are deleted. The fine poset has D as a minimum and three elements A, C, B above it. Hence its order complex is a cone and is contractible.

The bit is not isotone, so neither the possible nor forced map is isotone; no adjoint criterion applies. However every lower Quillen fiber is contractible: over D, A, C it is a point or edge, and over B it is the entire fine poset, again a cone. Therefore the refinement lies in

$$\mathbf{Q} \setminus \mathbf{A}.$$

11.4 Homotopy-neutral while both Quillen orientations fail

Take

$$\Phi = (X \wedge Y, X, X \vee Y).$$

On the four valuations 00, 01, 10, 11, the realized signatures form a four-element chain

$$p_0 < p_1 < p_2 < p_3$$

with

$$\begin{array}{l|l} 00 & \emptyset \\ 01 & \{X \vee Y\} \\ 10 & \{X, X \vee Y\} \\ 11 & \{X \wedge Y, X, X \vee Y\}. \end{array}$$

Add

$$\psi = \neg Y.$$

Its profile along the chain is

$$1, 0, 1, 0.$$

No coarse class splits. The surviving strict comparisons are

$$p_0 < p_2, \quad p_1 < p_2, \quad p_1 < p_3.$$

Thus O_Ψ is a three-edge tree, while O_Φ is a 3-simplex. Both are nonempty and contractible, so the refinement projection is a homotopy equivalence.

Nevertheless,

$$r^{-1}(\downarrow p_1) = \{p_0, p_1\}$$

is a two-point antichain, and

$$r^{-1}(\uparrow p_2) = \{p_2, p_3\}$$

is also a two-point antichain. Hence the lower-fiber Quillen criterion fails and the dual upper-fiber criterion fails. This gives a concrete refinement in

$$H \setminus Q.$$

11.5 Homology-active: adding negation to (X)

Let

$$\Phi = (X), \quad \Psi = (X, \neg X).$$

The coarse signature poset is a two-element chain, so O_Φ is an edge. The fine signatures $\{X\}$ and $\{\neg X\}$ are incomparable, so O_Ψ is two isolated points. Hence

$$H_0(O_\Phi) \cong \mathbb{F}_2, \quad H_0(O_\Psi) \cong \mathbb{F}_2^2.$$

The refinement is not homology-neutral. The mapping cone gives

$$D_1(\Psi/\Phi; \mathbb{F}_2) \cong \mathbb{F}_2, \quad D_0(\Psi/\Phi; \mathbb{F}_2) = 0.$$

This is the smallest example in which adding an observation splits a connected coarse information structure into disconnected fine states.

11.6 Splitting can still be homotopy-neutral

Let $\Phi = (X)$ on two variables X, Y , and add Y . Each coarse X -state splits into the two possible Y -values, so

$$\mathcal{B}_\emptyset = \mathcal{B}_{\{X\}} = \{0, 1\}.$$

The coarse poset is an edge and the fine poset is the Boolean diamond. Both are contractible. In fact both the possible map $M_Y \equiv 1$ and the forced map $m_Y \equiv 0$ are isotone, so the projection has both adjoints. This shows that state splitting by itself does not imply a topological defect.

12 Small exhaustive verification

To test the logical hierarchy independently of the proofs, the accompanying script exhaustively enumerates every one-bit refinement profile over every nonempty subposet of the Boolean square B_2 . A coarse point may have new-bit fiber $\{0\}$, $\{1\}$, or $\{0, 1\}$. There are 255 such refinement instances in total.

The verifier checks redundancy, left/right section-adjoint criteria, lower-fiber contractibility by finite beat-point reduction, lower-fiber acyclicity over $\mathbb{F}_2$, the Betti numbers of the coarse and fine order complexes, and the mapping-cone Betti numbers. It finds exactly five logical pattern types under those tests:

$(R, A, Q_\downarrow, \text{Acyc}_\downarrow, M)$	count
$(0, 0, 0, 0, 0)$	24
$(0, 0, 0, 0, 1)$	16
$(0, 0, 1, 1, 1)$	20
$(0, 1, 1, 1, 1)$	144
$(1, 1, 1, 1, 1)$	51

These finite checks are not substitutes for the theorems, but they verify the example calculations and demonstrate that the sufficient conditions are genuinely non-equivalent even at very small size.

13 Universality of refinement and the limit of complete homotopy classification

The preceding criteria invite the question whether one could push further to a complete effective homotopy classification. In unrestricted dimension the answer is negative.

Theorem 13.1 (Universality of refined signature posets). *[A]/[C] Every finite poset Q is isomorphic to the realized-signature poset of some finite propositional observation family. More strongly, Q can occur as P_Ψ in a refinement*

$$\Phi \subseteq \Psi$$

for which P_Φ is a single point.

Proof. Take enough propositional variables that $|\Omega_P| \geq |Q|$, and choose a surjection

$$h : \Omega_P \twoheadrightarrow Q.$$

Let the coarse observation family Φ be empty, so P_Φ is one point. Index new observations by the elements $q \in Q$. For each q , prescribe the truth set

$$T_q := \{v \in \Omega_P : q \leq h(v)\}.$$

Every subset of a finite valuation space is definable by a propositional formula, for example by the disjunction of the complete minterms of its members. Choose θ_q with $\varsigma(\theta_q) = T_q$.

If $h(v) = x$, then the new observation signature is

$$o_\Psi(v) = \{q \in Q : q \leq x\} = \downarrow x.$$

The map $x \mapsto \downarrow x$ is an order embedding of Q into 2^Q , because

$$x \leq y \iff \downarrow x \subseteq \downarrow y.$$

Surjectivity of h realizes every such signature and no others. Hence $P_\Psi \cong Q$. □

Corollary 13.2 (Every finite simplicial homotopy type occurs). *For every finite simplicial complex K , choose its face poset $Q = \mathcal{X}(K)$. Then*

$$\Delta(Q) = \text{sd}(K),$$

the barycentric subdivision of K . Therefore a refinement above a one-point coarse ProLT can realize the homotopy type of any finite simplicial complex.

Theorem 13.3 (Undecidability of unrestricted homotopy-neutral refinement). *[S]/[A]/[C] There is no algorithm that, for arbitrary finite ProLT observation refinements in unrestricted dimension, always decides whether*

$$\Delta(r_{\Psi\Phi})$$

is a homotopy equivalence.

Proof. Given a finite simplicial complex K , apply Corollary 13.2 to construct a refinement whose coarse signature poset is a point and whose fine order complex has the homotopy type of K . The refinement projection to the point is a homotopy equivalence exactly when K is contractible. Algorithmic recognition of contractibility for arbitrary finite simplicial complexes is undecidable. Thus a universal decision algorithm for homotopy-neutral ProLT refinements would decide an undecidable problem. $\square$

Corollary 13.4 (Homology remains effectively computable). *For any fixed computable coefficient field, homology-neutrality of a given finite refinement is decidable: construct the finite mapping-cone boundary matrices and compute their ranks. Thus the homotopy and homology classification problems diverge sharply in unrestricted generality.*

Remark 13.5 (Strictness of $H \subseteq M$). The universality theorem also imports standard acyclic noncontractible finite complexes. For example, a finite triangulation of a punctured Poincaré homology sphere is acyclic but has nontrivial fundamental group. Refining a point to its face poset gives a homology equivalence to the point which is not a homotopy equivalence.

14 A practical classification algorithm

The theory suggests the following release-quality workflow for a concrete observation refinement.

Step 1. Build the coarse poset. Compute the realized old signatures P_Φ and their inclusion order.

Step 2. Build fiber profiles. For each $p \in P_\Phi$, compute

$$\mathcal{B}_p = o_\Theta(C_p).$$

No fine-poset construction is conceptually necessary beyond these profiles.

Step 3. Check exact redundancy. If every $\mathcal{B}_p$ is singleton and the profile map is isotone, the refinement is an order isomorphism and the classification stops.

Step 4. Check adjoints. Compute $\mathcal{U}(p)$ and $\mathcal{L}(p)$. If all $\mathcal{U}(p) \in \mathcal{B}_p$, or all $\mathcal{L}(p) \in \mathcal{B}_p$, the refinement is homotopy-neutral.

Step 5. Check principal fibers. Apply beat-point/collapse tests or another contractibility certificate to $r^{-1}(\downarrow p)$ and $r^{-1}(\uparrow p)$. Uniform contractibility certifies simple homotopy equivalence. Uniform acyclicity certifies homology equivalence.

- Step 6. Compute the exact homology defect.** Build $\text{Cone}(r_{\#})$. Nonzero D_k proves that the refinement changes homology. Vanishing of all D_k proves homology equivalence.
- Step 7. For non-splitting refinements, use descent chains.** Replace the general mapping cone by the relative/descent chain complex of Proposition 6.3.
- Step 8. Do not treat an inconclusive homotopy test as failure.** The Quillen and adjoint criteria are sufficient, not necessary. In unrestricted dimension no complete decision algorithm exists. For a research theorem, restrict to a structurally controlled subclass.

15 Promising theorem targets after this classification

The classification above narrows the next research questions substantially.

- T1. Logical criteria for contractible Quillen fibers.** Find syntactic or semantic conditions on newly added formulas that force every $r^{-1}(\downarrow p)$ to have a cone point, beat-point reduction, or nerve decomposition.
- T2. Minimal homology-changing observations.** Given Φ , characterize or optimize observations ψ for which $D_k((\Phi, \psi)/\Phi) \neq 0$, possibly with a cost on formulas or observations.
- T3. Post- T_0 persistence.** Study the reverse filtration of Corollary 5.4. Determine whether logical properties of the added observations constrain barcode births/deaths in the relative descent complexes.
- T4. Refinement/update interaction.** Combine the contravariant-looking refinement deletion $O_{\Psi} \subseteq O_{\Phi}$ after T_0 with premise restriction subcomplexes. This should yield a two-parameter family indexed by observational resolution and admissible-state restriction.
- T5. Cross-layer comparison.** Relate refinement defects in canonical order-complex homology to changes in the presentation-sensitive Dowker compatibility complex and to changes in metric/Rips persistence. A nontrivial comparison theorem would be substantially stronger than defining the layers independently.
- T6. Restricted decidable classes.** Because unrestricted homotopy neutrality is undecidable, identify meaningful ProLT classes where the problem becomes decidable or efficiently certifiable: bounded poset height, bounded number of observations, distributive-lattice images, monotone formula fragments, or bounded treewidth of the comparability/Hasse graph.

16 Preliminary novelty positioning

The finite-poset fiber theorems, adjunction-homotopy principle, mapping cones, and undecidability of arbitrary finite-complex contractibility are standard mathematics. They should not be claimed as new. The current contribution candidates are instead the *ProLT-specific synthesis and sharpenings*: the refinement fiber-profile normal form; the exact redundancy theorem in signature terms; the possible/forced monotonicity criterion; the descent-chain description after T_0 ; the refinement-defect hierarchy; and the universality reduction formulated for propositional observation refinement.

A preliminary literature check found adjacent but distinct areas. Formal concept analysis and information-system theory study binary object–attribute data and attribute reduction. Separately, Gaucher used the phrase “refinement of observation” in directed homotopy theory for flows and finite bounded posets, with preservation

results for branching/merging homology and underlying homotopy type. That is not the same object as the ProLT signature-projection refinement studied here, but the terminology overlap should be acknowledged in any publication. A dedicated adversarial novelty audit remains necessary before promoting any [C] item to a novelty claim.

17 Conclusion

Observation refinement in a ProLT is not a single phenomenon. It may:

- add a coordinate without changing the topology at all;
- split observational states while preserving homotopy type;
- leave all states unsplit but delete specialization comparisons;
- change homotopy while preserving homology;
- or change homology itself.

The fiber-profile normal form makes these possibilities explicit. It reduces every refinement to a structured surjective poset map

$$Q_{\mathcal{B}} \longrightarrow P_{\Phi}.$$

Topological redundancy is exactly order isomorphism. Adjoint and Quillen-fiber tests give progressively broader homotopy-neutral classes. Mapping-cone homology gives an exact finite test for homology neutrality. Once the old ProLT is T_0 , refinement becomes especially transparent: it can no longer split vertices, so it acts only by deleting chains from the canonical order complex, and the defect is ordinary relative homology generated by the deleted descent chains.

The universality theorem also tells us where to stop looking for an all-purpose homotopy decision rule: unrestricted refinement already contains the full contractibility problem for finite complexes. The productive research direction is therefore not a mythical universal criterion, but a theory of *logically natural subclasses* in which the fiber geometry can be controlled and related back to observations, premises, and eventually time-evolving logical state spaces.

A Computational verification summary

The accompanying verifier constructs order complexes and normalized simplicial chain maps explicitly over $\mathbb{F}_2$. It verifies:

- (i) the redundant $X \wedge Y$ example has identical coarse/fine Betti vector $(1, 0, 0)$ and zero cone homology;
- (ii) the XOR refinement has coarse Betti $(1, 0, 0)$, fine Betti $(1, 0)$, and zero cone homology;
- (iii) the negation split has coarse Betti $(1, 0)$, fine Betti (2) , and cone Betti $(0, 1)$;
- (iv) the splitting-but-neutral Y refinement of (X) has zero cone homology;
- (v) the four-chain alternating profile $(1, 0, 1, 0)$ yields contractible coarse and fine complexes while failing both lower and upper Quillen-fiber tests;

- (vi) all 255 one-bit profiles over nonempty subposets of B_2 agree with the logical hierarchy reported in Section 10.

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